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CLASSIFICATION

Nearly 2 million kinds of animals have been identified by scientists, and thousands more are discovered every year. The real total may be 10 times this number, which means that the task of locating and identifying all the world's animals will never be complete. To make sense of this bewildering diversity, biologists use classification—a system that works like a gigantic filing system of all the past and present life on our planet. In classification, the entire living world is divided into groups, known as taxa. Each type of animal is given a unique scientific name, and is placed with its closest relatives, reflecting the path that evolution is thought to have followed.

Principles of classification

Scientific classification dates back over 300 years, to the work of John Ray (1627–1705). He classified plants and animals, using names that described features such as leaf shape. He also realized the importance of species, an idea that is still key to classification today. In the 18th century, another botanist—Carl Linnaeus (1707–1778)—laid the foundations of modern classification by devising classification hierarchies and 2-part scientific names. Like Ray, Linnaeus worked long before the discovery of evolution. He saw species as fixed, with unique shapes and lifestyles that were part of a divine plan.

At first sight, Linnaeus's binomial names may seem cumbersome, even though they are far more concise than the ones that were used before. They have 2 immense advantages. Unlike common names, they are unique to a particular species and they act like signposts, showing exactly where a species fits into life as a whole.

The study of classification, known as taxonomy, has rules for the way scientific names are devised and used. Each species has a genus name and a specific name by which it can be identified. The tiger, for example, is *Panthera tigris*, showing that its closest relatives are other members of the genus *Panthera*—the lion, the leopard, and the jaguar. The wild cat, on the other hand, belongs to the genus *Felis*, which includes most of the world's smaller cats. At a glance, this shows that there are differences between tigers and wild cats, in the way they have evolved as well as in their size.

Classification levels

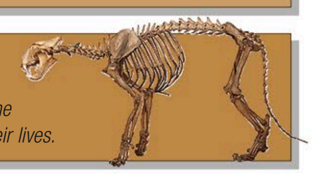
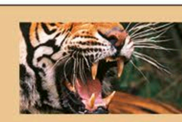
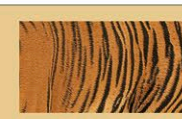
In Linnaean classification, the species level is the basic starting point. Species are organized into groups of increasing size, starting with genera and working upward through families, orders, classes, and phyla, and then into kingdoms and domains, which are the largest groups of all.

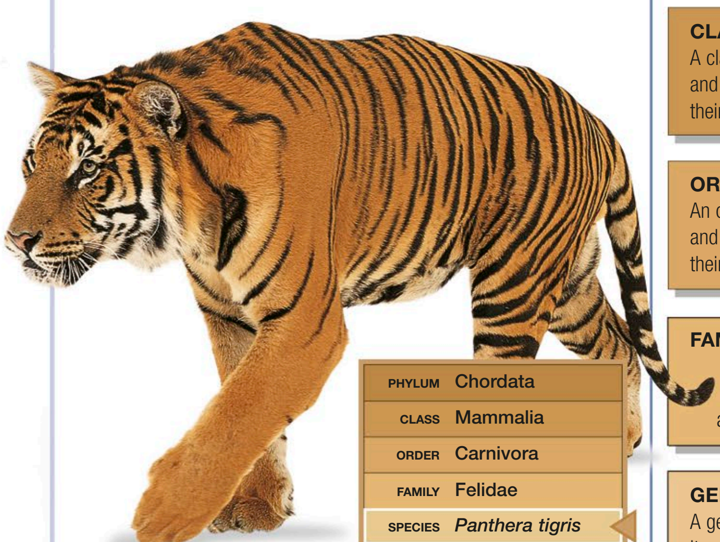
These groups work like flexible folders, and even equivalent levels can vary enormously in size. The cat family, for example, contains 37 species, while the aardvark family contains just one. At the other extreme, in the insect world, the weevil family currently contains 55,000 species, and there are probably many more. Not all animals fit neatly into groups that Linnaeus originally devised and so, over 2 centuries of study, taxonomists have created a wide array of intermediate levels. These include superfamilies, infraclasses, suborders, and subphyla, each fitting into the hierarchy at different points. At a much finer level, many species are divided into variants called subspecies—the milksnake, for example, has about 24, each with its own distinctive markings and geographic range.

These intermediate levels do not mean that Linnaeus's system does not work. Rather, it simply reflects the fact that classification levels are really convenient labels, rather than things that have a physical existence of their own. The only category that really exists is the species, and even this level can be difficult to define. Traditionally, a species is defined as a group of living things that breed exclusively with their own kind to produce fertile young. This works well enough with many animals, but does not always hold true.

Gathering evidence

Identifying which group an animal belongs to is vital to the classification process, and anatomical features play a key part. They can be very useful in tracing the path of evolution, often showing how body parts, such as limbs or teeth, have been developed and modified for different ways of life. The limbs of 4-legged vertebrates are one of the best-known examples of this kind of evolutionary improvization. The basic limb pattern, dating back

KINGDOM A kingdom is an overall division containing organisms that work in fundamentally similar ways.	Animalia *The kingdom Animalia contains multicellular organisms that obtain energy by eating food. Most have nerves and muscles, and are mobile.*	
PHYLUM A phylum is a major subdivision of a kingdom, and it contains one or more classes and their subgroups.	Chordata *Animals of the phylum Chordata have a strengthening rod or notochord running the length of their bodies, for all or part of their lives.*	
CLASS A class is a major subdivision of a phylum, and it contains one or more orders and their subgroups.	Mammalia *The class Mammalia contains chordates that are warm-blooded, have hair, and suckle their young. The majority of them give birth to live young.*	
ORDER An order is a major subdivision of a class, and it contains one or more families and their subgroups.	Carnivora *The order Carnivora contains mammals that have teeth specialized for biting and shearing. Many of them, including the tiger, live primarily on meat.*	
FAMILY A family is a subdivision of an order, and it contains one or more genera and their subgroups.	Felidae *The family Felidae contains carnivores with short skulls and well-developed claws. In most cases, the claws are retractable.*	
GENUS A genus is a subdivision of a family, and it contains one or more species and their subgroups.	***PANTHERA*** *The genus* Panthera *contains large cats that have a specialized larynx with elastic ligaments. Unlike other cats, they can roar as well as purr.*	
SPECIES A species is a group of similar individuals that are able to interbreed in the wild.	***PANTHERA TIGRIS*** *The tiger is the only member of the genus* Panthera *that has a striped coat when adult. There are several varieties, or subspecies.*	



PHYLUM	Chordata
CLASS	Mammalia
ORDER	Carnivora
FAMILY	Felidae
SPECIES	*Panthera tigris*

TIGER CLASSIFICATION

In this book, panels such as the one above are used to identify the position of animal groups in the taxonomic hierarchy. The larger panel on the right defines the various taxonomic ranks, starting at the top with the kingdom (one of the highest ranks) and ending with the species. Taking the tiger as an example, it also shows how a particular animal's physical characteristics are used to determine its place in the classification system.